

Codeforces and Polygon may be unavailable from [February 22, 6:30 \(UTC\)](#) to [February 22, 20:30 \(UTC\)](#) due to technical maintenance.

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS HACKS ROOM STANDINGS CUSTOM INVOCATION

D. Water Tree

time limit per test: 4 seconds
memory limit per test: 256 megabytes

Mad scientist Mike has constructed a rooted tree, which consists of n vertices. Each vertex is a reservoir which can be either empty or filled with water.

The vertices of the tree are numbered from 1 to n with the root at vertex 1. For each vertex, the reservoirs of its children are located below the reservoir of this vertex, and the vertex is connected with each of the children by a pipe through which water can flow downwards.

Mike wants to do the following operations with the tree:

- Fill vertex v with water. Then v and all its children are filled with water.
- Empty vertex v . Then v and all its ancestors are emptied.
- Determine whether vertex v is filled with water at the moment.

Initially all vertices of the tree are empty.

Mike has already compiled a full list of operations that he wants to perform in order. Before experimenting with the tree Mike decided to run the list through a simulation. Help Mike determine what results will he get after performing all the operations.

Input

The first line of the input contains an integer n ($1 \leq n \leq 500000$) — the number of vertices in the tree. Each of the following $n - 1$ lines contains two space-separated numbers a_i, b_i ($1 \leq a_i, b_i \leq n, a_i \neq b_i$) — the edges of the tree.

The next line contains a number q ($1 \leq q \leq 500000$) — the number of operations to perform. Each of the following q lines contains two space-separated numbers c_i ($1 \leq c_i \leq 3$), v_i ($1 \leq v_i \leq n$), where c_i is the operation type (according to the numbering given in the statement), and v_i is the vertex on which the operation is performed.

It is guaranteed that the given graph is a tree.

Output

For each type 3 operation print 1 on a separate line if the vertex is full, and 0 if the vertex is empty. Print the answers to queries in the order in which the queries are given in the input.

Examples

input	Copy
5 1 2 5 1 2 3 4 2 12 1 1 2 3 3 1 3 2 3 3 3 4 1 2 2 4 3 1 3 3 3 4 3 5	
output	Copy
0 0 0 1 0 1 0 1	

Codeforces Round 200 (Div. 1)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
363221998	Feb/16/2026 02:18	Accepted
363221727	Feb/16/2026 02:09	Wrong answer on test 6
363221488	Feb/16/2026 02:02	Wrong answer on test 6

→ Problem tags

data structures dfs and similar graphs trees *2100

No tag edit access

→ Contest materials

- Codeforces Round #200
- Tutorial